

Extra things

Definition: A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is periodic if there exists $T > 0$ such that $f(x+T) = f(x) \forall x$.

Definition: We say $f: D \rightarrow \mathbb{R}$ is monotonic increasing if $f(x) \leq f(y) \forall x < y \quad (x, y \in D(f))$

We have strictly monotonic increasing if $f(x) < f(y) \forall x < y$

Piecewise definitions of functions

Corresponds to splitting the domain into subsets with different formulae

Function operations

$$(f+g)(x) = f(x) + g(x), \quad D(f+g) = D(f) \cap D(g)$$

Same for others

$$f \circ g = f(g(x)), \quad D(f \circ g) = R(g)$$